ECE332 Lab 3: Waveguides

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Pre-lab assignment

1. Which of the following types of transmission lines does NOT support the propagation of transverse-electromagnetic (TEM) waves? (include all that apply.)

(a) Rectangular Waveguides

(b) Coaxial Cable

(c) Parallel-plate Lines

(d) Optical Fiber

2. What is the dominant mode for transverse magnetic (TM) waves propagating through a waveguide?

(a) TM_{00}

(b) TM_{10}

(c) TM_{01}

(d) TM_{11}

(e) None of the above

3. What is the dominant mode for the transverse electric (TE) waves propagating through a waveguide?

(a) TE_{00}

(b) TE_{01}

(c) TE_{10}

(d) TE_{11}

(e) None of the above

4. What type of filter does a resonant cavity implement?

(a) Low Pass

(b) High Pass

(c) Notch

(d) Bandpass

(e) None of the above

5. Which instrument is used for measuring the Standing Wave Ratio (SWR) if the signal is propagating through a waveguide?

(a) Systron/Donner 5000

(b) Hewlett Packard 415E

(c) Neither, you made those up!

Introduction

This lab investigates how electromagnetic waves travel inside rectangular waveguides, with a focus on propagation behavior, cutoff conditions, and the formation of standing wave patterns. Waveguides are hollow metal structures used to guide microwave-frequency signals, and unlike coaxial lines they cannot support TEM modes. Instead, they operate using transverse electric (TE) and transverse magnetic (TM) modes, each with a specific cutoff frequency determined by the waveguide's physical dimensions.

In this experiment, a slotted waveguide was used to visualize standing waves created by reflections from a short-circuit load. By sliding a probe along the slot, we measured changes in the electric field and identified the positions of maxima and minima. These measurements allowed us to determine the guide wavelength λ_g, as well as calculate phase and group velocities at different operating frequencies. Overall, the lab emphasizes how boundary conditions and waveguide geometry shape wave propagation and how these effects differ from ordinary free-space transmission.

Part 1: Wavelength Measurements

Objective

The goal of this experiment is to study how electromagnetic waves travel inside a rectangular waveguide and to measure the guide wavelength (λ_g) at several microwave frequencies. Using a slotted waveguide and an SWR meter, we observe the standing wave pattern created by a short-circuit termination. From the measured minima spacing, we calculate the guide wavelength, phase velocity (v_p), and group velocity (v_g) for the dominant TE_{10} mode.

Theory

A rectangular waveguide is a hollow metal structure that carries electromagnetic waves only when the operating frequency is above a certain cutoff value. For the dominant TE_{10} mode, the cutoff wavelength is $\lambda_c = 2a/m$ where a is the wider dimension of the waveguide and $m = 1$ for the TE_{10} mode.

The relationship between the guide wavelength λ_g, the free-space wavelength λ, and the cutoff wavelength λ_c is:

$$\lambda_g = \frac{\lambda}{\sqrt{1 - (\lambda/\lambda_c)^2}}$$

Inside the waveguide, the wave travels with a phase velocity and group velocity given by:

$$v_p = \frac{c}{\sqrt{1 - (f_c/f)^2}}, \qquad v_g = c\sqrt{1 - (f_c/f)^2}$$

Here f_c is the cutoff frequency and f is the operating frequency. Their product remains constant: $c^2 = v_p v_g$. This means that as the phase velocity becomes greater than the speed of light, the group velocity (which carries energy) decreases at the same rate, keeping the physics consistent.

Setup

The experiment uses a slotted rectangular waveguide connected to a Systron-Donner 5000 oscillator operating from 8–12.4 GHz. A movable probe inside the slotted line detects the electric field intensity, and an HP 415E SWR meter displays the measured signal.

Steps:

1. Attach a short-circuit plate to the end of the waveguide.
2. Connect the probe output to the SWR meter using a BNC cable.
3. Set the oscillator to the desired test frequency (e.g., 11 GHz).
4. Adjust the probe depth to get a strong reading without disturbing the field.
5. Slide the probe along the slot to find the maxima and minima of the standing wave.
6. Record the positions of at least three minima or maxima to calculate $\lambda_g/2$.

Equations Used

Waveguide Wavelength: $\lambda_g = 4\Delta d = 4(d_{n+1} - d_n)$
Phase Velocity: $v_p = f\lambda_g$
Group Velocity: $v_g = c^2/v_p$, where $c = 3 \times 10^8$ m/s (speed of light in vacuum)

Results

$f = 11$ **GHz**				
	Position (mm)	λ_g (mm)	v_p (m/s)	v_g (m/s)
Min 1	170.1	31.2	3.432×10^8	2.622×10^8
Max 1	162.3	36.0	3.960×10^8	2.273×10^8
Min 2	153.3	32.4	3.564×10^8	2.525×10^8
Max 2	145.2	34.4	3.784×10^8	2.378×10^8
Min 3	136.6	29.6	3.256×10^8	2.764×10^8
Max 3	129.2	N/A	N/A	N/A
Avg:		32.7	3.599×10^8	2.501×10^8

$f = 9.5$ **GHz**				
	Position (mm)	λ_g (mm)	v_p (m/s)	v_g (m/s)
Min 1	174.4	44.0	4.180×10^8	2.153×10^8
Max 1	163.4	43.6	4.142×10^8	2.173×10^8
Min 2	152.5	42.0	3.990×10^8	2.256×10^8
Max 2	142.0	42.4	4.028×10^8	2.234×10^8
Min 3	131.4	43.6	4.142×10^8	2.173×10^8
Max 3	120.5	N/A	N/A	N/A
Avg:		43.1	4.096×10^8	2.197×10^8

$f = 8.5$ **GHz**				
	Position (mm)	λ_g (mm)	v_p (m/s)	v_g (m/s)
Min 1	166.4	62.4	5.304×10^8	1.697×10^8
Max 1	150.8	47.6	4.046×10^8	2.224×10^8
Min 2	138.9	64.4	5.474×10^8	1.644×10^8
Max 2	122.8	46.0	3.910×10^8	2.302×10^8
Min 3	111.3	58.0	4.930×10^8	1.826×10^8
Max 3	96.8	N/A	N/A	N/A
Avg:		55.7	4.733×10^8	1.902×10^8

The tables above show the waveguide SWR minimums and maximums.

Measured and recorded dimensions of a and b of the waveguide:
a (width) = 22.92 mm, b (height) = 10.29 mm.

Discussion

From the measurements, the guide wavelength (λ_g) becomes shorter as the operating frequency increases. For example, λ_g decreased from about 55.7 mm at 8.5 GHz to 32.7 mm at 11 GHz. This agrees with waveguide theory, which predicts that higher frequencies produce shorter wavelengths inside the guide. The calculated phase velocities v_p are greater than the speed of light, which is normal for waveguides because phase velocity does not represent the speed of energy flow. Meanwhile, the group velocity v_g (the actual energy transfer speed) remains below c and gets smaller as frequency decreases. This opposite behavior between v_p and v_g satisfies the condition $v_p v_g = c^2$. Overall, the results match the expected behavior of a rectangular waveguide operating in the TE_{10} dominant mode. Minor differences in the measured maxima and minima could be due to probe positioning errors or small inaccuracies in the SWR meter, but the overall trends remain consistent with theory.

Part 2: Frequency Measurements

Objective

The goal of this part of the experiment is to measure the resonance frequencies of electromagnetic waves in a rectangular waveguide using a tunable cavity. By finding the frequencies at which the cavity resonates, we can identify the operating frequencies of the dominant TE_{10} mode and verify the relationship between frequency, wavelength, and wave propagation velocities.

Theory

Electromagnetic waves can only propagate through a rectangular waveguide if their frequency is above the cutoff frequency. For the dominant TE_{10} mode, the cutoff frequency is $f_c = c/(2a)$ where a is the larger dimension of the waveguide and c is the speed of light. The phase velocity and group velocity for a wave traveling in the guide are given by:

$$v_p = \frac{c}{\sqrt{1-(f_c/f)^2}}, \qquad v_g = c\sqrt{1-(f_c/f)^2}$$

At resonance, the tunable cavity absorbs the most power, which shows up as a sharp dip on the SWR meter. Measuring this resonance frequency allows us to confirm the expected propagation behavior of the waveguide and compare it with theory.

Setup

This setup is similar to Part 1, but now a tunable cavity (HP X532B frequency meter) is placed between the slotted waveguide carriage and the short-circuit termination.

Steps:

1. Remove the short-circuit termination from the carriage and attach it to the output of the tunable cavity.
2. Connect the frequency meter to the slotted waveguide while keeping the oscillator at 11 GHz.
3. Adjust the oscillator frequency and slowly tune the cavity until a sharp dip appears on the SWR meter, indicating resonance.
4. Read the resonant frequency directly from the frequency meter scale.
5. Repeat the measurements for 9.5 GHz and 8.5 GHz inputs.

Equations Used

$f_c = c/(2a)$, where $c = 3 \times 10^8$ m/s (speed of light in vacuum) and $a = 22.92$ mm (width of the waveguide)

Phase Velocity: $v_p = \dfrac{c}{\sqrt{1-(f_c/f_{\text{measured}})^2}}$

Group Velocity: $v_g = c\sqrt{1-(f_c/f_{\text{measured}})^2}$ (equivalently $v_g = c^2/v_p$)

Results

Oscillator f (GHz)	Measured f (GHz)	v_g (m/s)	v_p (m/s)
11	11.050	2.489×10^8	3.616×10^8
9.5	9.405	2.219×10^8	4.055×10^8
8.5	8.515	1.898×10^8	4.741×10^8

Table 2: Resonant cavity frequencies.

The measured resonance frequencies were slightly lower than the oscillator settings for two of three cases, which is expected due to small losses and calibration limits in the setup. As the operating frequency decreased, the group velocity also decreased, while the phase velocity increased, matching waveguide theory.

The results confirm the relationship $v_p v_g = c^2$, showing that energy travels more slowly near cutoff, while phase velocity becomes greater than the speed of light (without violating physics, since phase velocity does not represent energy flow).

Overall, the measurements matched the expected behavior of the TE_{10} mode, with small variations likely caused by probe alignment or tuning sensitivity.

Part 1 Questions

1) Why are λ and λ_g different?

In free space, a wave moves in a straight line, so its wavelength is just $\lambda = c/f$. Inside a waveguide, the wave keeps bouncing off the metal walls and travels in a zig-zag path. Because the wave travels a longer path, its effective wavelength becomes larger. This new wavelength is called the guide wavelength λ_g, and it is always longer than the free-space wavelength for the same frequency.

2) What electric field would the probe detect at the short-circuit end?

At the short-circuit end of a waveguide, the electric field must be zero because the metal walls force the tangential electric field to be zero. So if the probe could touch the shorted end, it would measure $\mathbf{E} = \mathbf{0}$. This happens because the boundary condition at a perfect conductor requires the tangential electric field at the surface to be zero.

3) Expected distance to the nearest detectable minimum (and comparison)

The short circuit forces $E = 0$ at $z = 0$. Minima repeat every $\lambda_g/2$ along the guide. The nearest detectable minimum is $n \times (\lambda_g/2)$ from the termination, where n is the smallest integer that places the minimum within probe reach.

f (GHz)	$\lambda_g/2$ (mm)	n	Expected d (mm)	Measured d (mm)	% diff
11	16.36	5	81.80	83.5	2.1%
9.5	21.56	4 ($\approx$ 4.43)	86.24	95.6	10.9%
8.5	27.84	3	83.52	85.4	2.3%

The 11 GHz and 8.5 GHz results agree within about 2%, consistent with typical vernier scale reading error. The 9.5 GHz result shows a larger discrepancy (about 10%), likely due to probe alignment or a measurement made to a different minimum than expected.

4) Number of half-wavelengths between the nearest minimum and the short-circuit end

Using $n = d_{\text{measured}}/(\lambda_g/2)$:

f (GHz)	Measured d (mm)	$\lambda_g/2$ (mm)	$n = d/(\lambda_g/2)$	Nearest int n
11	83.5	16.36	5.10	5
9.5	95.6	21.56	4.43	4 (with uncertainty)
8.5	85.4	27.84	3.07	3

This matches what we expect for the TE_{10} standing-wave pattern, where the electric field is zero at the short and reaches its first minimum a half-wavelength away.

5) Using a and b to compute cutoff modes

(a) From the cutoff equation:

$$f_c = f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

In free space: $1/\sqrt{\mu\varepsilon} = c = 3 \times 10^8$ m/s. Therefore,

$$f_c = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \times 10^8}{2}\sqrt{\left(\frac{m}{22.92 \text{ mm}}\right)^2 + \left(\frac{n}{10.29 \text{ mm}}\right)^2}$$

Mode	(m, n)	f_c formula	f_c (GHz)
TE_{10}	$(1, 0)$	$c/(2a)$	6.54 (lowest)
TE_{20}	$(2, 0)$	c/a	13.09
TE_{01}	$(0, 1)$	$c/(2b)$	14.58
TE_{11}/TM_{11}	$(1, 1)$	$(c/2)\sqrt{1/a^2 + 1/b^2}$	15.98 (4th, not in top 3)

Three lowest: TE_{10} (6.54 GHz) $<$ TE_{20} (13.09 GHz) $<$ TE_{01} (14.58 GHz). Note $a/b = 22.92/10.29 = 2.23 \approx 2$, confirming standard X-band ordering $TE_{10} < TE_{20} < TE_{01}$.

(b) Propagating modes (need $f > f_c$):

- 8.5–11 GHz $>$ 6.54 GHz $\Rightarrow$ TE_{10} propagates
- 8.5–11 GHz $<$ 13.09 GHz $\Rightarrow$ TE_{20} does NOT propagate
- 8.5–11 GHz $<$ 14.58 GHz $\Rightarrow$ TE_{01} does NOT propagate

Only TE_{10} propagates. The guide operates single-mode throughout.

(c) Back-calculating f_c from measured λ_g. Starting from $v_p = f\lambda_g = c/\sqrt{1 - (f_c/f)^2}$ and solving for f_c:

$$f_c = f\sqrt{1 - \left(\frac{c}{f\lambda_g}\right)^2} = f\sqrt{1 - \left(\frac{\lambda}{\lambda_g}\right)^2}$$

Frequency (GHz)	λ (mm)	λ_g (mm)	f_c (GHz)
11	27.27	32.73	6.08
9.5	31.58	43.12	6.47
8.5	35.29	55.68	6.57

Average back-calculated f_c = 6.37 GHz vs geometric f_c (TE_{10}) = 6.54 GHz, giving a percent difference of 2.6%. These values all match the theoretical TE_{10} cutoff within about 3%. Small variations come from probe accuracy and λ_g averaging.

Part 2 Questions

Q1

Using the measured values for frequency and wavelength, calculate the phase velocity of the wave: $v_p = \lambda_g f$. How close (i.e. percentage difference) is this to your values from Experiment 3.1? Based on this, how important is it to know the frequency accurately?

Frequency (GHz)	v_p (m/s) (Part 1)	v_p (m/s) (Part 2)	% Difference
11	3.599×10^8	3.616×10^8	0.5%
9.5	4.096×10^8	4.055×10^8	1.0%
8.5	4.733×10^8	4.741×10^8	0.2%

The phase velocities from Part 2 agree with Part 1 to within 1.1%, confirming that the oscillator frequency was close to its set value and that the waveguide wavelength measurements were accurate. Small discrepancies are attributable to vernier scale reading error and minor oscillator frequency drift.

Based on this, knowing the frequency accurately is important but not critical at this level. Since $v_p = \lambda_g f$, a 1% error in frequency directly produces a 1% error in v_p. More critically, frequency accuracy matters for mode control: if the operating frequency drifted above the TE_{20} cutoff (13.09 GHz) it could excite unwanted higher-order modes and corrupt measurements. The cavity meter confirms the oscillator stays within about 0.5% of its set value, which is sufficient for single-mode waveguide operation.

Q2

What happens to the group velocity v_g as frequency gets closer to the cutoff frequency f_c? Can you offer some physical explanation for this?

From Eq. 4: $v_g = c\sqrt{1-(f_c/f)^2}$. As $f \to f_c$, the term $(f_c/f)^2 \to 1$, so $\sqrt{1-(f_c/f)^2} \to 0$ and therefore $v_g \to 0$. The measured data confirms this trend:

f (GHz)	f_c (GHz)	f/f_c	v_g (m/s)
11	6.54	1.68	2.501×10^8
9.5	6.54	1.45	2.197×10^8
8.5	6.54	1.30	1.902×10^8

As frequency decreases toward $f_c = 6.54$ GHz, v_g drops from 2.501×10^8 to 1.902×10^8 m/s, approaching zero at cutoff.

Physically, near cutoff the wave stops transporting energy efficiently. Most of its energy stays stored in the electric and magnetic fields instead of moving forward. Because of this, the wave travels very slowly down the guide, and energy flow almost stops. Visualizing the TE_{10} mode as two plane waves bouncing obliquely between the broad walls at angle θ to the z-axis (where $\sin\theta = f_c/f$), as $f \to f_c$ the angle $\theta \to 90^\circ$ and the waves bounce nearly perpendicular to the axis with almost no net axial advance. Below cutoff the mode becomes evanescent and decays exponentially.